

Industry 4.0 Key Research Topics: A Bibliometric Review

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Abstract—The digitalization phenomenon and the growing adoption of innovative technologies is favoring the transformation of organizations into Cyber Physical Systems. The transition towards this new organizational system is often labeled as Industry 4.0. The attention to this topic is increasing lately, even if heterogeneous denominations and approaches are adopted to address this phenomenon. This article aims to shed some light between twelve different approaches of industry 4.0 described in literature. Conducting a bibliometric review, the paper identifies the key topics related to Industry 4.0 by the literature available.

Keywords—industry 4.0; smart industry; smart manufacturing; review

I. INTRODUCTION

Nowadays a lot of attentions are posed to Industry 4.0 [1] and its impact on the organizations' life, by both scholars and practitioners. Industry 4.0 could be defined as the integration of technologies (i.e. Big Data Analytics [2], Cloud Services [3], 3D-Printing [4], Cyber Security [5], Autonomous Robots [6], Internet of Things [7], Augmented Reality [8], Simulation [9], Horizontal and Vertical Integration [10]), which allows the transformation of how organizations operate along with high changes in business models and manufacturing processes. This phenomenon originally spread out from the Hannover Conference in 2011 and it is often labeled as the 4th industrial revolution. The adoption of the above-mentioned set of technologies enable the deep transformation of organizations into Cyber Physical System (CPS) i.e. "systems of collaborating computational entities which are in intensive connection with the surrounding physical world and its on-going processes, providing and using, at the same time, data-accessing and data-processing services available on the internet" [11]. Inside this CPS, the workers' role will be completely different from the traditional one. CPS will require additional workers executing supervising tasks. Routinized jobs will disappear as these activities are going to be replaced by the machine automation. Even though scholars and practitioners underline the huge impact of the implementation of these technologies, the terms and the approaches adopted to address this phenomenon usually change from country to country. The most relevant approaches are briefly synthesized below.

- *Smart Manufacturing* is a label generally used as synonym of industry 4.0 with no link to any specific government plan.
- *Smart Industry* is promoted by the Dutch Government as an Action Agenda that "aim [...] to make the industry more competitive through faster and better utilization of the opportunities ICT has to offer".
- *Fabbrica Intelligente* is supported by the Italian Government; its goals are the development of a strategy based on research and innovation, able to lead to competitive advantage of Italians manufacturing companies, and it could also transform the products, the processes and the productive systems of those companies.
- *Nouvelle France Industrielle* is the French way of addressing Industry 4.0; its focus is on the re-industrialization, leading French companies in the adoption of net technologies and in the changing of the business models.
- *Smart Factory or Manufacturing Innovation 3.0* is a South Korean set of policies that should allow companies to pursue the computerization of the manufacturing processes.
- *Made in China 2025* is a Chinese public plan with the ultimate goal of pushing traditional manufacturing companies to embrace the new digital technologies.
- *Make in India* identifies a set of incentives offered to manufacturing companies, especially for Research and Development department, by the Indian Government.
- *Catapult* is developed by the UK Government; its main objective is to strengthen UK manufacturing companies and allow them to embrace innovation.
- *Industrial Value Chain Initiative (IVI)* is sponsored by the Japanese Government and it is also known as "Connected! Manufacturing". This initiative "Actively discusses how human-centric manufacturing will change with IoT, IVI aims at building a mutually connected system architecture, based on collaboration areas between companies".
- *Produktion 2030* is the Swedish's Government program that aim to strengthen the competitiveness of Swedish manufacturing firms, through the adoption

of new knowledge, new technology and good partnership.

- *Industria Conectada* is a set of public policies, established by the Spanish Government, which aims in developing new skills for industry 4.0 jobs. It also promotes multidisciplinary collaboration and the implementation of the industry 4.0 technologies.
- *Prumysl 4.0* is defined by the Chzech Government as a set of incentive plans for those companies who decide to invest in the digital transformation.
- *Advanced Manufacturing Partnership*, also known as Advanced Manufacturing 2.0, is the national program of the United States of America. Its ultimate goal is to increase the general competitiveness of the USA through the creation of high valuable jobs and securing the leadership of US manufacturing companies in the adoption of new technologies.

Given this heterogeneity of labels, the authors decide to carry out a bibliometric analysis of the field including all the different denominations listed above (Table I).

TABLE I. KEYWORDS PARAMETERS

Keywords Searched	“Advanced Manufacturing Partnership”; “Catapult”; “Fabbrica Intelligente”; “Industria Conectada”; “Industrial Value Chain Initiative”; “Made in China 2025”; “Make in India”; “Nouvelle France Industrielle”; “Produktion 2030”; “Prumysl 4.0”; “Smart Industry”; “Smart Factory”; “Smart Manufacturing”;
Database	Scopus
Time Limitations	None
Category Limitations	Business, Management, Accounting
Source	Peer reviewed Article
Language	English
Date of query	November 2017
Number of documents	117

II. METHODOLOGY

The authors investigate such heterogenous field of research through a bibliometric study. This methodology was chosen because it allows to process large numbers of data and to identify the patterns between them. Lastly, it supports the identification of the most investigated research trends.

The above list of the most important labels of Industry 4.0 is used to perform a query on the Scopus database. Then the analysis is conducted according to the workflow created by [12], this workflow consists of five steps:

- *Research Design*: Identification of the research question/s and choice of the proper bibliometric method;
- *Compilation of bibliometric data*: Selection of the most suitable database for conducting the query;

- *Analysis*: The elaboration of the data in a bibliometric software;
- *Visualization*: Choice of the visualization method;
- *Interpretation*: illustration of the results.

In particular, to investigate this field the authors adopted a co-word analysis technique [13], which analyzes the co-occurrence of the keywords. The co-occurrence [14] of a keyword is the number of time a keyword appears together with others in a document. The more they appear together the stronger is the link between them. A set of closely related keywords forms themes (or cluster) of research [15]. The analysis produces a map as an output that gives a clear definition of the topics composing the research field under investigation.

The analysis is carried out using a software named VosViewer [16]. This software handles different bibliometric methods (e.g. co-citation analysis, co-authorship) and it is recognized as really accurate in doing a co-word analysis. The network analysis of the identified data supported the design of the three main maps listed below.

1) *Network Visualization* (Fig. 1): It displays circles, the bigger the circle is, the more weight that item possess.

2) *Density Visualization*: The more the area is red the more dense the area is, which indicates a great number of items in that zone.

3) *Overlay Visualization*: Circle in this map are colored according to the average year of publications.

In order to ensure the reliability of this literature review, only peer reviewed articles published on journal are taken in to consideration. Due to the international nature of the resulting documents, the English language of the articles is another limitation posed. Since the purpose of this research is to identify the topic discussed in literature of industry 4.0 from an organizational perspective, and since this field of research attracts a lot of attentions from the engineering scholars, the research is limited to the categories of Business, Management and Accounting. As a consequence, only relevant literature related to the topic of interest is selected.

III. FINDINGS

As shown in Table I, the research identifies 117 documents, and 618 keywords associated with these documents. This keywords selection is used to conduct the analysis. Table II shows the twenty most important keywords that emerges from the bibliometric analysis. The table also show the occurrences of the keywords and the total link strength [17]. The Occurrence index show how many times a keyword appears in the documents object of analysis. The total link strength indicates how good are the connection between a certain keyword to the others.

Fig. 1 shows the keywords emerging from the analysis. According to Table II, the most important are: Industry 4.0, Manufacture, Internet of Things, Cyber Physical System, Supply Chain Management and Human Resource Management. Despite the definition of nine main clusters of keywords, the keywords appear to be very close.

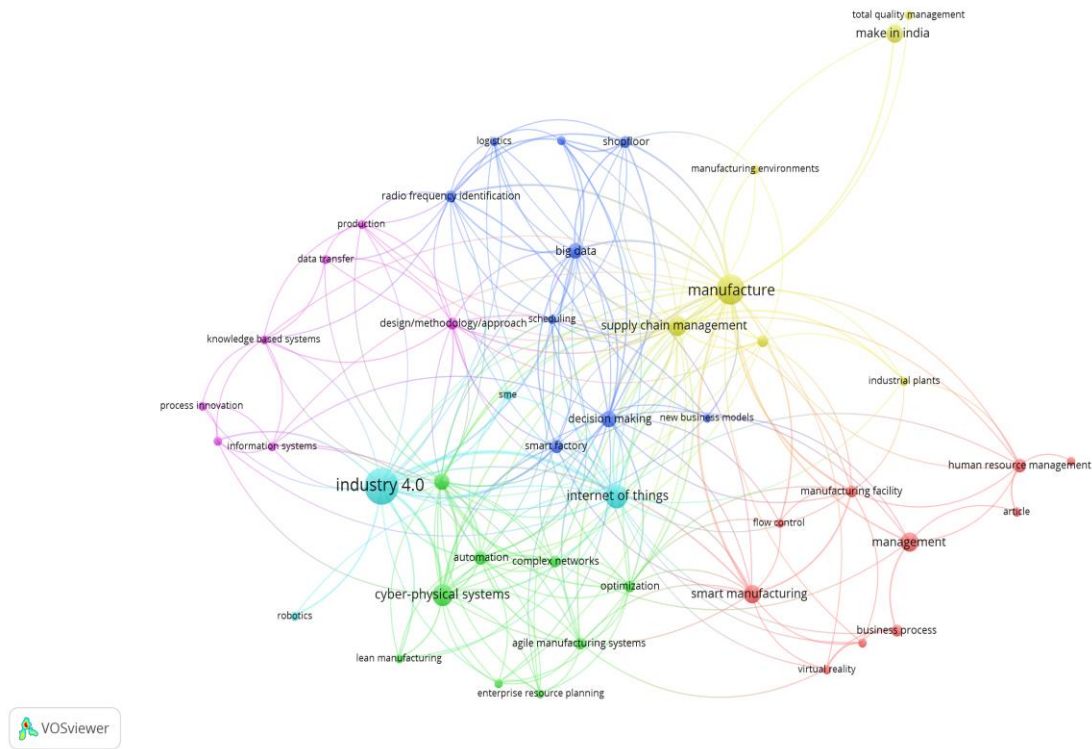


Figure 1. Network Visualization

TABLE II. KEYWORDS OCCURRENCES

Keywords	Occurrences	Total link strength
Industry 4.0	28	47
Manufacture	19	62
Internet of Things	12	41
Cyber physical systems	10	34
Supply Chain Management	9	39
Management	8	9
Smart Manufacturing	7	19
Make in India	7	4
Decision Making	6	32
Embedded Systems	6	28
Big Data	5	25
Automation	4	22
Smart Factory	4	17
Human Resource Management	4	10
Radio Frequency Identification	3	20
Optimization	3	19
Design/Methodology/approach	3	16
Shopfloor	3	16
Agile Manufacturing Systems	3	15
Industrial revolutions	3	14

This support the idea that despite the different labels the topic that compose the literature around industry 4.0 are the same, since almost the totality of the keywords are close to each other.

The only topic that is clearly distant from the other ones is the one related to the “Make in India” initiative. This topic, despite being connected with the hot-topic of Manufacture, does not connect well with the others. This is mainly because this Indian government plan is composed by several policies and it does not put much attention in the digitalization of the production processes. Actually, the only close keyword in the map to “Make in India” is the one about “Total Quality Management”.

Industry 4.0 is the most important topic along with the Manufacture one. Researches directly investigating the Industry 4.0 topic focused on different aspects, this testify the multifaced nature of this theme. [18] finds Industry 4.0 to be in an initial stage. [19] identifies a system architecture for Industry 4.0, while [20] showed that this innovation could harm companies especially SMEs: the smaller the companies the more critical their survival could be. The Manufacture topic is also closely related to the Supply Chain Management, since many articles have tried to improve the supply chain of manufacturing companies, that now are facing the challenges posed by Industry 4.0, which is supposed to heavily change the Supply Chain Management [21] especially since this implementation produce shift, among all, in the processes’ speed [22].

Furthermore, it is worth to notice the almost complete absence from the map of the different labels listed in Table I. This highlight the fact that, despite the differences in the name used to refer to the phenomenon called “industry 4.0”,

the discussion inside the literature is on similar topics, which underlie that the object of the research is usually the same.

A certain degree of importance is generated by the topics which involve a system perspective: Lean Manufacturing, Agile Manufacturing, Enterprise Resource Planning. These keywords are strongly associated to Industry 4.0, and their disposition on the map is between Cyber Physical System, Smart Manufacturing and Internet of Things, which indicates how central these topics are. In particular [23] identifies that “committing into Industry 4.0 makes a factory lean besides being smart”, while [24] sees Industry 4.0 as the possibility to implement the Lean Automation in the factories.

Lastly, it is interesting to note that almost all the keywords are new, as the Industry 4.0 terms appeared for the first time in 2013. Only two keywords are older than the others. The first one is Data Transfer, which relates to the Radio Frequency Identification (RFID). This is a key component of the digitalization of the manufacturing companies [25] as it allows the transformation of normal object or products into smart ones [26].

The other keyword is “Human Resource Management” (HRM). This is a key topic and it is recognized by several authors to play an important part in enforcing the successful implementation of the industry 4.0. Its association with Manufacture and Supply Chain Management (Fig. 1) remind the importance of this organizational function in assuring the good implementation of other organizational model (e.g. Lean Management, Green Management, Environmental Management). Moreover, HRM is also recognized crucial in supporting the improvement of the operational processes in manufacturing enterprises. This is also confirmed by the study of [27]. The authors underline the relevance of HR flexibility and its impact on organizational innovation and culture.

The key role played by the HR department, is also noticeable by the worries that surrounds the Industry 4.0 topic. In November 2017, the consulting firm McKinsey highlights in its report that almost 800 million of jobs are at risk due to the implementation of new technologies. This huge change will push for the automation, expelling from the market jobs with almost no added value. However, this technologies’ implementation is creating a skills shortage which bring the HR department to choose whether is beneficial for companies to acquire those needed skills on the market or to train the old employees from the inside. On this topic a research of [28] shows the importance of overtraining (in advanced manufacturing environment) to ensure high quality apprenticeships of new personnel. There is also room for concern due to the “job polarization” that could be created by industry 4.0 [29]. The job polarization is the creation of two kind of works: one with routinized tasks and the others more focused on problem solving. According to this view, the low skilled task will be replaced by automation.

The last finding that emerges from the analysis of the literature on the Industry 4.0, is the lack of theory connected to this topic. The researchers adopt a practitioners’ perspective more than a theoretical one. This lack of theory supporting Industry 4.0 serves as another clue in

understanding how fragmented is the field in this beginning stage.

IV. DISCUSSIONS AND CONCLUSIONS

This article investigates the research field of the Industry 4.0. In order to identify the more discussed topics inside this field, a list of the most important labels used to address this phenomenon is created, so that it could be possible to get strong and essential data supporting an effective literature review.

The heterogeneity of the labels identified reflects the conspicuous amount of governmental action supporting the technologies’ implementation. Therefore, the methodology chosen for investigate the field is a bibliometric technique. This methodology is recognized as effective in supporting a co-word analysis. In this kind of technique, the unit of analysis are the keywords that appears in title, abstract and keywords section of every paper. The analysis is based on the 618 authors’ keywords of the selected papers, the outcomes are objectives and not based on subjective interpretation of the authors. The outputs are synthesized in two tables (Table I, Table II) and one figure (Fig. 1) showing the connection between the keywords.

Results from this analysis suggest the closeness of the topic discussed inside this literature, which means that, despite the differences in the adopted labels by the different countries to address Industry 4.0, the key elements covered by researchers are the same. Lastly, Industry 4.0 is a complex and disruptive. The articles inside this field analyze different aspects, from the change in the production processes to the skills shortage that will need HRM support, along with the fears of the job losses.

V. LIMITATIONS

While this research provides a useful support in understanding the Industry 4.0 literature, there is room for improvement, and possible future directions for articles. In order to secure the goodness of the results only peer reviewed articles were included in the analysis. This choice left out some documents from the analysis. A systematic literature review on this “leftover” could improve the strength of the results that emerges from this research. Since there is a time gap between the submission on a journal and its publication. Including those results could help the identification of the most recent topic discussed inside the industry 4.0 literature.

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